

BEYOND AVERAGING: ALL-ROUNDER CAPABILITY EVALUATION FOR RADIOLOGY REPORT GENERATION

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ABSTRACT

Radiology Report Generation (RRG) holds great promise for reducing radiologists’ workload and standardizing clinical practice. The existing evaluation protocols largely rely on aggregating metrics over the entire test set from averaging perspective. However, due to the inherent imbalance of medical imaging datasets across clinically meaningful subgroups (e.g., age, gender, disease type, comorbidities, imaging view), previous evaluating strategies by aggregating metrics averagely fail to capture performance heterogeneity over those subgroups. As a result, such evaluations may misrepresent a model’s true capability for RRG. To address this issue, we propose **All-Rounder Capability Evaluation (ARCE)**, a novel evaluation paradigm inspired by subgroup analysis in clinical research. ARCE partitions the test set into clinically relevant subgroups and systematically integrates model performance across them, thereby quantifying the ability of models to maintain robust performance under diverse clinical conditions. Extensive experiments on the large-scale public dataset MIMIC-CXR demonstrate that ARCE effectively identifies “specialized” models that achieve high aggregate scores yet generalize poorly, while providing more reliable model rankings. Our findings highlight ARCE as a practical and fair framework for selecting models that consistently perform well across heterogeneous clinical scenarios.

Index Terms— Radiology Report Generation, Evaluation Paradigm, Subgroup Analysis, Robustness, Model Selection

1. INTRODUCTION

Radiology Report Generation (RRG) aims to automatically interpret medical images and generate structured, clinically accurate reports, thereby reducing workload and improving diagnostic efficiency [1, 2, 3, 4]. Recent advances in deep learning—particularly large language models and vision-language models—have driven notable progress in RRG[5, 6].

However, the existing evaluation practices for RRG face a critical limitation, which we term the **evaluation bottleneck**. Conventional metrics—including language-oriented measures (e.g., BLEU[7], ROUGE-L[8]) and clinically-oriented measures (e.g., CheXbert F1[9, 10], RadGraph F1[11])—are all Marco Aggregate Metrics (MAM) from simple averaging perspective over entire test set. As illustrated in Fig. 1(a), MAM computes a single score across the entire test set, masking subgroup disparities caused by inherent data imbalance. The RRG models may thus achieve high aggregate scores by mastering frequent cases while failing on rare but clinically important ones. While MAM are useful for coarse benchmarking, they fail to reflect a model’s true generalization ability for RRG.

We have released the complete code and data split at [Link].

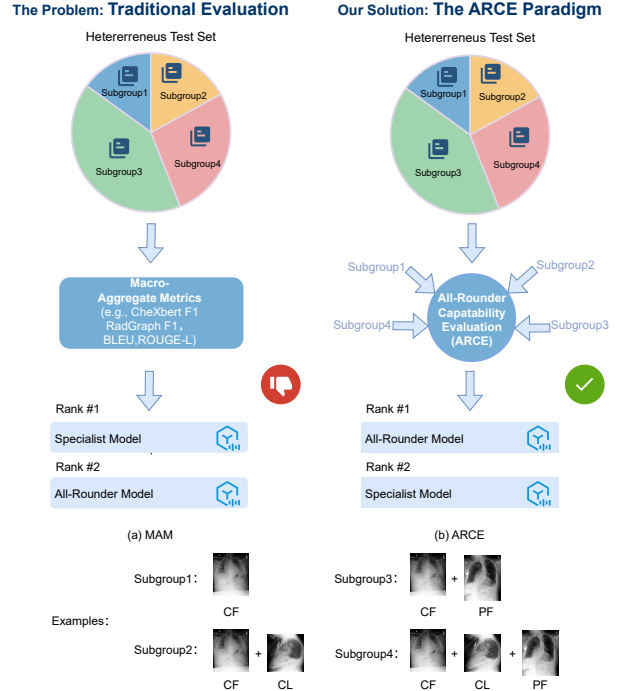


Fig. 1. Compared to MAM, our ARCE can obviously capture the imbalance gap by dicriminating different subgroups.

Despite the issue of data imbalance in RRG[12], no prior work has systematically evaluated models across clinically meaningful subgroups, which is a key gap this paper addresses. As shown in Fig. 1(b), we propose the **All-Rounder Capability Evaluation (ARCE)**, which not only introduces subgroup analysis explicitly, but also perform fine-grained metric aggregation. By jointly assessing subgroup mean performance and fairness (via Gini coefficient), ARCE rewards models that demonstrate robustness across diverse clinical scenarios, enabling more reliable and clinically faithful evaluation. In summary, our contributions are threefold:

1. We propose the **first subgroup-based evaluation framework** for RRG, grounded in clinically meaningful stratification.
2. We introduce the **ARCE strategy**, which balances mean performance and fairness across subgroups to quantify generalization ability.
3. We demonstrate **ARCE’s flexibility** by applying it to subgrouping based on disease rarity, showing its generalizability as a scalable evaluation tool.

2. RELATED WORKS

Evaluating Radiology Report Generation (RRG) is crucial for advancing this field. Existing metrics can be grouped into two categories. Natural language generation (NLG) metrics (e.g., BLEU, ROUGE-L, METEOR) [13, 14] focus on n-gram overlaps but ignore clinical semantics [15]. Clinical efficacy (CE) metrics (e.g., CheXbert F1, RadGraph F1, or LLM-based evaluators like RadFact[3]) directly assess factual correctness and have become mainstream. However, all these metrics typically produce a single aggregated score over the entire test set, which is highly sensitive to data imbalance and tends to reward “specialist” models that perform well in frequent scenarios but poorly in rare, clinically important cases.

Attempts to mitigate imbalance include Micro F1 (dominated by frequent labels) and Macro F1 (treating labels equally), but these remain confined to the label level. In RRG, heterogeneity arises not only from label frequency but also from diverse clinical scenarios, such as multi-view interpretation or longitudinal comparisons, which are overlooked by these approaches[16].

In clinical research, subgroup analysis is a standard practice to uncover treatment effects across heterogeneous populations. Researchers can explore whether a treatment is more effective for a specific population by dividing the patient population into subgroups based on characteristics such as age, sex, disease severity, or biomarkers[17, 18]. Inspired by this principle, we introduce the ARCE framework, which integrates subgroup-based evaluation into RRG. ARCE moves beyond aggregate scores by explicitly quantifying robustness and fairness across clinically meaningful subgroups, offering a more reliable benchmark for real-world deployment.

3. METHODOLOGY

3.1. Preliminaries and Task Definition

The radiology report generation (RRG) task takes as input a set of radiographic images $\mathbf{X}$ (which may contain one or multiple views) and outputs a textual report $\mathbf{G}$, which should be factually and clinically consistent with the reference report $\mathbf{R}$ authored by radiologists. A standard evaluation metric can be formulated as a function $\mathcal{F}(G, R) \mapsto \text{Score}$, mapping the generated report and the reference report to a real-valued score. In practice, these scores are typically aggregated over the test set (e.g., by averaging). However, if the test set consists of multiple unevenly distributed subgroups, then the aggregated metrics will be misleading due to the excessive weight of the majority group.

3.2. Defining Clinically Meaningful Subgroups

The cornerstone of our evaluation paradigm is the stratification of the test set into clinically meaningful, mutually exclusive subgroups. The partitioning chosen for this study is based on the types of imaging views included in each radiology examination, which directly reflects the diversity of clinical workflows and the richness of diagnostic information:

CF (Current Frontal only): The most basic and common scenario, reflecting the model’s ability to handle basic scenarios.

CF+CL (CF + Current Lateral): Lateral views provide orthogonal anatomical information, enabling better localization and detection of findings obscured in frontal views[19].

CF+PF (CF + Prior Frontal): Longitudinal comparison is a core part of radiological practice, allowing assessment of interval

changes (e.g., disease progression or resolution)[20].

CF+PF+CL (Comprehensive views): The most information-rich and diagnostically challenging setting, requiring multi-view integration and temporal reasoning.

These subgroups are constructed from the MIMIC-CXR dataset metadata (see Sec. 4.1). Importantly, they capture clinically realistic diversity while exposing potential weaknesses of models that rely heavily on majority cases.

3.3. All-Rounder Capability Evaluation (ARCE)

Given N disjoint subgroups $\mathcal{D}_{test} = \mathbf{D}_1 \cup \mathbf{D}_2 \cup \dots \cup \mathbf{D}_N$, we compute subgroup-specific scores $\{s_i\}_{i=1}^N$ for a chosen MAM $\mathcal{F}$ (e.g., CheXbert F1, RadGraph F1, or other CE indicators):

$$s_i \triangleq \mathcal{F}(G_i, R_i), \quad i = 1, \dots, N. \quad (1)$$

We then introduce the *All-Rounder Capability Evaluation (ARCE)* score, defined as

$$ARCE \triangleq \mu \cdot (1 - \mathbf{G}), \quad (2)$$

where μ is the mean performance across subgroups and $\mathbf{G}$ is the Gini coefficient of the subgroup scores. Formally, μ is defined as:

$$\mu \triangleq \frac{\sum_{i=1}^N w_i s_i}{\sum_{i=1}^N w_i}, \quad (3)$$

where w_i denotes the size (or relative importance) of subgroup D_i . In our research, we set all weights $w_i = 1$ to treat each subgroup equally. Intuitively, μ reflects the overall ability of the model. This design ensures that ARCE does not only emphasize fairness but also rewards strong overall performance. In this way, models that are uniformly poor across subgroups (e.g., [0.14, 0.15, 0.15, 0.14]) will not be mistakenly regarded as superior simply due to their fairness.

Additionally, the calculations for $\mathbf{G}$ is:

$$\mathbf{G} \triangleq \frac{\sum_{i=1}^N \sum_{j=1}^N w_i w_j |s_i - s_j|}{2\mu \sum_{i=1}^N \sum_{j=1}^N w_i w_j}. \quad (4)$$

Here, $\mathbf{G}$ quantifies inequality across subgroups. A lower $\mathbf{G}$ indicates more consistent performance, while a higher $\mathbf{G}$ reflects substantial disparities. Specifically, $\mathbf{G} = 0$ corresponds to perfectly equal subgroup performance, whereas $\mathbf{G} = 1$ represents extreme imbalance where some subgroup scores approach zero. This penalization mechanism prevents models that achieve high scores only on dominant subgroups—but fail on rare yet clinically important ones—from being overvalued.

Note that we adopt the Gini coefficient as the fairness indicator rather than standard deviation. This is because the goal of ARCE is not to measure statistical *dispersion* but *inequality*. Standard deviation, which measures deviation from the mean, captures volatility but lacks a notion of fairness. In contrast, the Gini coefficient originates from economics as a measure of resource allocation inequality, making it more aligned with the notion of subgroup fairness. Moreover, Gini offers three practical advantages over variance-based penalties: (i) it is scale-invariant and normalized in $[0, 1]$, ensuring comparability across metrics. In contrast, standard deviation is an absolute metric. Its value varies with the dimensions of the data. For example, the standard deviation of a set of performance scores [80, 80, 90, 90] is different from that of set [0.8, 0.8, 0.9, 0.9]. (ii) it is based on pairwise subgroup differences rather than deviations from a central mean, better capturing disparities among subgroups; and (iii) it

is less sensitive to outliers, emphasizing systematic rather than incidental imbalance. However, standard deviation is very sensitive to extreme values because it includes square terms in its calculation. If the performance on a subset crashes drastically, it will disproportionately increase standard deviation.

Rationale Comparison with Traditional Weighted Average.

It is important to emphasize that our method is fundamentally different from another common strategy for handling subgroup imbalance, i.e., weighted average¹ as the final score. This Weighted averaging is a **priori and data-driven compensation**, addressing the question: “If certain subgroups are deemed more important, what should the overall score be?” By contrast, our ARCE is a **posteriori and performance-driven evaluation**, focusing on whether a model performs fairly and consistently when all tasks are considered equally important. In short, *weighted averaging corrects for dataset imbalance, whereas ARCE evaluates imbalance in model performance.*

Another key distinction lies in the treatment of weights. Weighted averaging requires predefined weights—most often inverse frequency weighting—which remain fixed once chosen. ARCE, in contrast, does not rely on preset weights. Its fairness component, the Gini coefficient, is dynamically derived from the model’s actual subgroup performance [21], directly quantifying inequality. Weighted averaging, however, has no explicit mechanism to measure such disparities [22].

Furthermore, weighted averaging inherently rewards models that perform well on high-weight subgroups, making it essentially similar in nature to aggregated scores. ARCE avoids this issue by rewarding models that achieve both strong and balanced performance across all subgroups.

Finally, weight selection represents the major weakness of weighted averaging: there is no universally objective standard for setting weights. Different application scenarios may call for different weighting schemes, inevitably introducing arbitrariness. ARCE elegantly sidesteps this problem, since the Gini coefficient is entirely determined by observed model performance, without requiring any subjective prior assumptions. *The detailed experiments can refer to section 4.2.*

4. EXPERIMENTATION

To validate the effectiveness of the proposed evaluation paradigm, we conduct a series of comprehensive experiments. This section introduces the experimental setup, presents the core results, and provides both quantitative and qualitative analyses to support our claims.

4.1. Experimental Setup

Dataset. All experiments are conducted on the publicly available MIMIC-CXR-JPG v2.0.0 dataset, one of the largest and most widely used benchmarks for paired chest X-ray images and radiology reports. We strictly follow the official train/validation/test split to ensure reproducibility.

To construct subgroup samples, we leverage the *ViewPosition* and *ViewCodeSequence_CodeMeaning* fields in the metadata to determine the imaging view type. For each CF image, we create an individual sample. If a lateral view exists within the same study (*study.id*), we randomly select one as the corresponding lateral view. To incorporate prior images, we track all studies of the same patient

¹Note that our RRG is not a classification problem and cannot directly apply traditional sampling strategies or imbalanced losses.

Table 1. Baseline Performance with traditional MAM Evaluation.

MAM	BLEU1	Rouge.L	neg_Macro_14.F1	neg_Micro_14.F1
MODEL ₁ ↑	27.47	22.03	28.15	45.15
MODEL ₂	27.22	20.86	27.23	43.34
MODEL ₃	27.42	19.61	27.18	43.68

Table 2. Subgroup Analysis of Model Performance

Macro/Micro_14	MODEL ₁	MODEL ₂	MODEL ₃
CF	20.13/35.54	22.77/39.28	23.06/39.75
CF+CL	20.99/33.23	21.04/33.14	21.06/30.85
CF+PF	29.73/50.90	27.25/47.87	27.92/48.49
CF+PF+CL	25.78/38.97	27.29/38.76	26.48/38.43

using *subject.id*, and identify earlier studies by comparing *Study-Date*. The most recent PF image is then paired with the current study. If multiple prior frontal images are available, multiple corresponding samples are constructed. Following this procedure, each sample can be categorized into one of the four clinical subgroups defined in Section 3.2. More details are given in the Appendix.

Unless otherwise noted, in tables we highlight the best and worst model with blue and red, respectively.

4.2. Results and Discussion

We present a detailed analysis of our experimental results to assess the effectiveness of our evaluation framework. The analysis is organized around four aspects: baseline comparison, subgroup analysis, re-ranking with the proposed ARCE score, and robustness analysis. For clarity, we report results in a series of tables and provide analysis below each. In image processing, MODEL₁ follows a LLaVA-style [23] direct concatenation strategy, while MODEL_{2,3} adopt an expert-based fusion strategy, conceptually related to METransformer [24]. Details are provided in the Appendix.

Baseline Comparison. Table 1 reports the performance of four models on MAM. To ensure comparability, we independently tuned hyperparameters such as effective batch size, epochs, and text input format so that all models achieved similar scores on conventional metrics. From these results, one might conclude that MODEL₁ performs best, and that MODEL₂ and MODEL₃ are difficult to distinguish.

Ablation Study. To examine subgroup-specific behavior, Table 2 presents model performance across clinically meaningful subgroups. The results reveal substantial discrepancies: the same model may perform well on certain subgroups while failing on others. Combined with Table 1, we can see conventional metrics often fail to capture subgroup disparities, models exhibit strong specialization that can mislead evaluation under imbalanced test sets.

Model Re-ranking with Our Proposed Strategy. Table 3 presents the evaluation results obtained with our proposed ARCE strategy. In contrast to MAM, ARCE establishes a clearer ranking by explicitly penalizing subgroup inequality. Notably, MODEL₁, which previously appeared strongest under aggregated metrics, is substantially downgraded due to its poor generalization across subgroups. This demonstrates ARCE’s ability to reveal “specialized” models that mask weaknesses under traditional evaluation.

Ground-Truth Validation on a Balanced Set. Finally, to validate ARCE against ground truth, we constructed a subgroup-balanced subset of 2400 samples from unused MIMIC data and re-evaluated all models. Table 4 shows that the ranking derived from ARCE aligns closely with that on the balanced set, confirming

Table 3. Evaluation with Our Proposed Strategy ARCE

ARCE	BLEU1	Rouge.L	neg_Macro_14_F1	neg_Micro_14_F1
MODEL ₁ ↓	26.58	21.37	22.06	36.13
MODEL ₂ ↑	26.58	20.47	23.13	36.97
MODEL ₃	27.04	19.46	23.13	35.99

Table 4. Validation on a Balanced Test Set(input views)

Selection Criteria	BLEU1	Rouge.L	neg_Macro_14_F1	neg_Micro_14_F1
MODEL ₁ ↓	19.84	17.79	12.01	19.83
MODEL ₂ ↑	21.48	17.44	13.35	23.02
MODEL ₃	21.69	17.00	12.53	21.59

that ARCE reliably approximates the performance ordering under unbiased conditions.

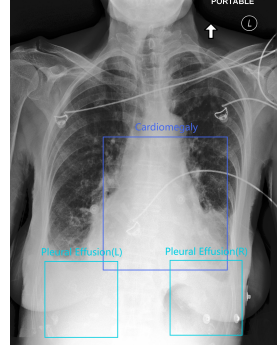
Overall, these results confirm that ARCE is an effective tool for identifying truly generalizable models in radiology report generation.

Comparison with Weighted Averaging. Table 5 reports weighted average scores calculated with three common strategies. In this table, we can see that when the weights are equal, the method can barely manage to identify the best model, but it fails to distinguish which of the other two models is better. With prevalence weights (reflecting the true test set distribution), the ranking changes again, incorrectly favoring MODEL₁, this is clearly inconsistent with our results. Using inverse-frequency weights, the method identifies MODEL₁ as the worst, but the relative ranking of MODEL₂ and MODEL₃ remains unclear. These results confirm our discussion in Section 3.3, showing that the weighted strategy does not actually measure the fairness of the model. The weighted strategy is highly sensitive to external, human-determined weighting schemes, leading to inconsistent and sometimes misleading conclusions. Overall, weighted averaging provides only vague comparisons and cannot reliably reflect true model quality, making it an unreliable evaluation approach.

Case Study. Here we show the difference in the results of the selected models on the test set. The highlighted part is the lesion reflected in this paragraph of text. Blue is correct and red is wrong. More cases are given in the Appendix.

4.3. Generality of the Proposed Paradigm

To further demonstrate the generality of our proposed paradigm, we conducted an additional set of experiments. In the main study, subgroup division was primarily based on the diversity of input views. However, in real-world clinical applications, another factor of particular interest is the rarity of certain pathologies. Therefore, in this section, we additionally stratify the dataset according to the prevalence of pathological findings, forming subgroups of common versus rare conditions. Notably, we keep the training data unchanged, further demonstrating the generality of our evaluation framework. According to the results of Table 6 and 7, this complementary evaluation aims to verify that the effectiveness of our method extends beyond view diversity and also effective in pathology-based subgrouping, thereby supporting the robustness and applicability of the proposed approach.



Ground Truth: Small bilateral pleural effusions are present. There is no focal consolidation. The lungs are hyperinflated with emphysematous changes. Heart size is increased.

MODEL₁: The heart size is top normal in configuration, unchanged since the prior study. There are diffuse patchy opacities bilaterally, concerning for pneumonia or aspiration. Mild pulmonary edema is also noted. No large pleural effusion seen.

MODEL₂: The right hemidiaphragm is elevated with air under the apex. There are multiple areas of consolidation in the left lung, including bilateral pleural effusions and possible pulmonary edema. No pneumothorax or pneumomediastinum. Heart size is enlarged.

Fig. 2. Case study on the CF subgroup. Image: input frontal chest X-ray. Text: generated reports from different models compared with the ground truth.

Table 5. The result of weighted average of different weights

equal/prevalence/inv	MODEL ₁	MODEL ₂	MODEL ₃
BLEU1	27.36/27.47/27.06	27.24/27.22/27.13	27.66/27.42/27.72
Rouge.L	22.51/22.02/22.69	21.18/20.85/21.33	20.28/19.61/20.76
neg_Macro_14_F1	24.16/26.92/21.55	24.59/26.47/22.76	24.63/26.51/22.87
neg_Micro_14_F1	39.66/44.01/35.91	39.76/42.72/37.39	39.38/42.74/36.73

Table 6. Sub-analysis of Model Performance on Disease Rarity

Macro/Micro_14	MODEL ₁	MODEL ₂	MODEL ₃
RARE	11.37/12.90	13.46/15.97	16.60/18.87
COMMON	24.51/48.50	23.97/47.07	24.43/47.76

Table 7. Evaluation with the Proposed Strategy on Disease Rarity

ARCE	BLEU1	Rouge.L	neg_Macro_14_F1	neg_Micro_14_F1
MODEL ₁ ↓	27.09	21.81	14.66	21.8
MODEL ₂	27.32	20.8	16.09	23.74
MODEL ₃ ↑	27.28	19.43	18.56	26.02

5. CONCLUSION

In this work, we reveal a fundamental limitation in current RRG evaluation: an over-reliance on MAM, which can systematically favor “specialized” models that perform well on frequent, simple cases but fail in rare and complex ones. To address this, we propose a clinically informed subgroup evaluation paradigm and introduce the **All-Rounder Capability Evaluation (ARCE)**, which jointly accounts for overall performance and subgroup fairness.

Despite these contributions, our study is limited by the scope of subgroup definitions and reliance on a single dataset. Future work includes extending ARCE-aware optimization into training objectives, exploring richer subgroup definitions (e.g., demographics, pathology types), and applying the paradigm across modalities and domains such as CT, MRI, or pathology image analysis.

Overall, this work provides not only a practical evaluation tool but also a shift in perspective: moving from maximizing a single score toward developing AI systems that are robust, fair, and clinically trustworthy.

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